

Go Further

David Moreno

Multiple dataframes

R allows you to work with multiple data objects (dataframes, lists, matrices, etc). When you work on a project sometimes you have two or three databases that have common observations. For example, in a clinical trial, you have one excel with patient codes and survival data and another file that contains patient codes and immunohistochemistry or mutations data. When the databases are very large or you are responsible of the project, manually merging databases is not a good practice.

Dplyr (tidyverse) has a friendly way to deal with these databases.

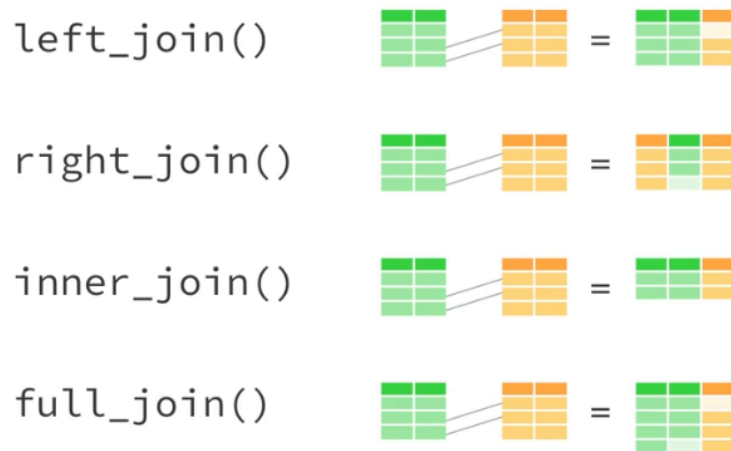


Image source: Screenshot from [R Pubs Tutorial by William Surles](#). Used for educational purposes.

```
# Work with the myeloma data and add PRDM1 mutations data
library(readxl)
library(tidyverse)
myeloma <- read_excel("/Users/dfmoreno/Dropbox/Lectures/myeloma.xlsx")
prdm1 <- read_excel("/Users/dfmoreno/Dropbox/Lectures/data/myeloma_prdm1.xlsx")
myeloma_2 <- read_excel("/Users/dfmoreno/Dropbox/Lectures/data/myeloma_2.xlsx")

# Check the joining
left <- left_join(myeloma, prdm1, by = "Patient")
right <- right_join(myeloma, prdm1, by = "Patient")
inner <- inner_join(myeloma, myeloma_2, by = "Patient")
full <- full_join(myeloma, myeloma_2, by = "Patient")
```

Modify values

There are multiple tools to modify values, feel free to explore other options in R.

```
# Keep working with the myeloma database
myeloma %>%
  rename(chr1 = chr1q21_status,
         Patient_Id = Patient)

myeloma %>%
  mutate(molecular_group_new = ifelse(molecular_group == "Hyperdiploid", 1, 0))

myeloma %>%
  mutate(molecular_group_new = case_when(molecular_group == "Hyperdiploid" ~ 1,
                                         TRUE ~ 2))

myeloma %>%
  mutate(molecular_group_new = case_when(molecular_group == "Hyperdiploid" ~ 1,
                                         molecular_group == "MMSET" ~ 3,
                                         TRUE ~ 2))

myeloma %>%
  mutate(molecular_group = case_when(molecular_group == "Hyperdiploid" ~ 1,
                                     TRUE ~ 2))
```

You can also modify multiple names of columns.

```
# Work with the "diagnoses" data
library(irr)
```

```
data("diagnoses")

diagnoses %>%
  select(rater1)

diagnoses %>%
  select(matches("rater"))

gsub("rater", "physician", colnames(diagnoses))

diagnoses %>%
  rename_with(~ gsub("rater", "physician", .))
```

Use substitutions

```
# Work with the both "diagnoses" and myeloma data
gsub("^", "test_", colnames(diagnoses)) # beginning
gsub("$", "_test", colnames(diagnoses)) # end
gsub("[1-9]", "x", colnames(diagnoses)) # any number
gsub("[a-z]", "A", colnames(diagnoses)) # each character
gsub("\\s", "_", myeloma$chr1q21_status) # empty space
gsub("\\s", "_", myeloma$molecular_group) # empty space
sub("\\s", "_", myeloma$molecular_group) # empty space
```

Reproducibility

Why is it important to keep track of all the work we do? Now, please check Figure 1B from this [paper](#) and try to reproduce exactly the results.

First, you should go to the “Data availability” section of the manuscript and click on the [repository](#). Then go to the folder 2 (mutation burden analyses), click on the analyses SNV mixed model, download the html file and open it. You will also need the data, which is located in the same folder (colonyinfo_AX00...).

```
# install.packages("nlme")
# Run exactly what it says in the first part
library(nlme)

sig_prop_type <- read.table("/Users/dfmoreno/Downloads/colonyinfo_AX001_KX001_KX002_KX003_TX
                           header=T, stringsAsFactors = F, sep="\t")
```

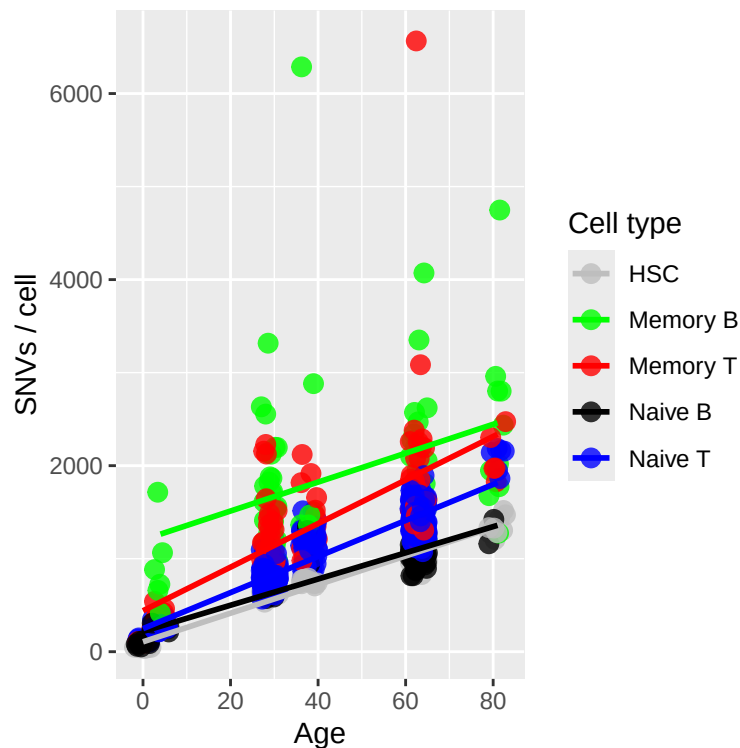
```

sig_prop_type$AgeFactor <- factor(sig_prop_type$Age, levels=c(81,63,38,29,4,0))
m3Allsub <- subset(sig_prop_type, Cell.type2 != "Treg")
full.lymph.Age_cell.interaction.lme <- lme(fixed = Nmut_hsc_as ~ Age * Cell.type2,
                                           random = ~ 1 | Donor / Cell.type2,
                                           weights = varIdent(form= ~ 1 | Cell.type2),
                                           data = m3Allsub, method="ML")

summary(full.lymph.Age_cell.interaction.lme)

# This is not the exact plot of the manuscript
# Further analyses are lacking due to time
ggplot(m3Allsub, aes(x = Age, y = Nmut_hsc_as, colour = Cell.type2)) +
  geom_jitter(alpha = 0.8, width = 2, size=3) +
  geom_smooth(method="lm", se = F) +
  scale_color_manual(
    name = "Cell type",
    values = c("grey", "green", "red", "black", "blue")) +
  xlab("Age") +
  ylab("SNVs / cell")

```



We reproduced in part the somatic mutation burden analysis results using code provided by Machado et al. (2022) available at [github](#)

This work is licensed under a

[Creative Commons Attribution-ShareAlike 4.0 International License](#).

